Dietary composition in the treatment of polycystic ovary syndrome: a systematic review to inform evidence-based guidelines.

While lifestyle management is recommended as first-line treatment of polycystic ovary syndrome (PCOS), the optimal dietary composition is unclear. The aim of this study was to compare the effect of different diet compositions on anthropometric, reproductive, metabolic, and psychological outcomes in PCOS. A literature search was conducted (Australasian Medical Index, CINAHL, EMBASE, Medline, PsycInfo, and EBM reviews; most recent search was performed January 19, 2012). Inclusion criteria were women with PCOS not taking anti-obesity medications and all weight-loss or maintenance diets comparing different dietary compositions. Studies were assessed for risk of bias. A total of 4,154 articles were retrieved and six articles from five studies met the a priori selection criteria, with 137 women included. A meta-analysis was not performed due to clinical heterogeneity for factors including participants, dietary intervention composition, duration, and outcomes. There were subtle differences between diets, with greater weight loss for a monounsaturated fat-enriched diet; improved menstrual regularity for a low-glycemic index diet; increased free androgen index for a high-carbohydrate diet; greater reductions in insulin resistance, fibrinogen, total, and high-density lipoprotein cholesterol for a low-carbohydrate or low-glycemic index diet; improved quality of life for a low-glycemic index diet; and improved depression and self-esteem for a high-protein diet. Weight loss improved the presentation of PCOS regardless of dietary composition in the majority of studies. Weight loss should be targeted in all overweight women with PCOS through reducing caloric intake in the setting of adequate nutritional intake and healthy food choices irrespective of diet composition.